

## 1. Purpose

This spike validates Paged.js pagination capabilities for the DocMaker architecture decision. The purpose paragraph is intentionally long enough to interact with pagination behaviour in a realistic way. It repeats: controlled laboratory documentation requires deterministic page-oriented rendering with headers, footers, page numbering, and repeated table headers across page boundaries.

### 1.1 Background

Background paragraph one. Quality manuals, standard operating procedures and work instructions all rely on stable pagination. This paragraph pads the page so the following table begins near the page boundary, which is required to observe multi-page table behaviour.

Background paragraph two with additional filler content so that the table below starts close to the bottom of page one. Repeat filler: accreditation documentation, calibration records, environmental monitoring, corrective action registers, internal audit schedules and management review minutes.

Table 1 — Sample Acceptance Criteria

| Parameter | Method           | Limit        | Result | Remark |
|-----------|------------------|--------------|--------|--------|
| pH        | IS 3025 Pt. 40   | 6.5 – 8.5    | 7.2    | Pass   |
| Turbidity | IS 3025 Pt. 21   | < 1 NTU      | 0.4    | Pass   |
| TDS       | IS 3025 Pt. 26   | < 500 mg/L   | 312    | Pass   |
| Chloride  | IS 3025 Pt. 23   | < 250 mg/L   | 88     | Pass   |
| Hardness  | IS 3025 Pt. 19   | < 200 mg/L   | 142    | Pass   |
| Iron      | IS 3025 Pt. 59   | < 0.3 mg/L   | 0.11   | Pass   |
| Fluoride  | IS 3025 Pt. 60   | < 1.0 mg/L   | 0.42   | Pass   |
| Sulphate  | IS 3025 Pt. 30   | < 200 mg/L   | 96     | Pass   |
| Nitrate   | IS 3025 Pt. 34   | < 45 mg/L    | 7.9    | Pass   |
| Colour    | IS 3025 Pt. 2    | < 15 TCU     | 4      | Pass   |
| Odour     | Organoleptic     | Acceptable   | Nil    | Pass   |
| EC        | IS 3025 Pt. 302  | < 1500 µS/cm | 640    | Pass   |
| Calcium   | IS 3025 Pt. 45   | < 100 mg/L   | 58     | Pass   |
| Magnesium | IS 3025 Pt. 45   | < 100 mg/L   | 29     | Pass   |
| Sodium    | IS 3025 Pt. 47   | < 200 mg/L   | 64     | Pass   |
| Potassium | Flame photometry | < 50 mg/L    | 12     | Pass   |
| Zinc      | AAS              | < 5.0 mg/L   | 0.08   | Pass   |
| Copper    | AAS              | < 1.5 mg/L   | 0.03   | Pass   |
| Manganese | AAS              | < 0.1 mg/L   | 0.02   | Pass   |
| Lead      | AAS              | < 0.01 mg/L  | < 0.01 | Pass   |

| Parameter                  | Method             | Limit        | Result  | Remark |
|----------------------------|--------------------|--------------|---------|--------|
| Arsenic                    | Hydride generation | < 0.01 mg/L  | < 0.01  | Pass   |
| Cadmium                    | AAS                | < 0.003 mg/L | < 0.003 | Pass   |
| Nickel                     | AAS                | < 0.02 mg/L  | 0.005   | Pass   |
| Chromium                   | AAS                | < 0.05 mg/L  | 0.01    | Pass   |
| Cyanide                    | IS 3025 Pt. 42     | < 0.05 mg/L  | < 0.02  | Pass   |
| Phenol                     | 4-AAP              | < 0.001 mg/L | < 0.001 | Pass   |
| MBAS                       | IS 3025 Pt. 15     | < 0.2 mg/L   | 0.05    | Pass   |
| Residual Cl                | DPD                | < 0.2 mg/L   | 0.1     | Pass   |
| Coliform                   | MPN                | Absent       | Absent  | Pass   |
| E. coli                    | MPN                | Absent       | Absent  | Pass   |
| Total plate count          | IS 3025 Pt. 90     | < 100 CFU/mL | < 1     | Pass   |
| Halogens                   | Titration          | < 200 mg/L   | 121     | Pass   |
| Alkalinity                 | Titration          | < 200 mg/L   | 150     | Pass   |
| Free CO <sub>2</sub>       | Titration          | Report       | 8       | Info   |
| Oil & grease               | Hexane extract     | < 10 mg/L    | < 1     | Pass   |
| SS                         | Gravimetric        | < 100 mg/L   | 22      | Pass   |
| BOD                        | IS 3025 Pt. 58     | < 30 mg/L    | 4       | Pass   |
| COD                        | Reflux titration   | < 100 mg/L   | 18      | Pass   |
| Ammonia                    | Phenate            | < 0.5 mg/L   | 0.12    | Pass   |
| Hardness CaCO <sub>3</sub> | Titration          | < 200 mg/L   | 142     | Pass   |
| Moisture                   | Karl Fischer       | Report       | 0.04%   | Info   |
| Conductivity               | IS 3025 Pt. 302    | < 1500       | 640     | Pass   |
| Particle count             | ISO 11500          | Report       | Class 8 | Info   |
| Viscosity                  | Brookfield         | Report       | 1.01 cP | Info   |
| Flash point                | Pensky-Martens     | Report       | N/A     | Info   |
| Density                    | pycnometer         | Report       | 0.998   | Info   |
| Melting point              | Capillary          | Report       | N/A     | Info   |
| Assay                      | Titration          | 99.5%        | 99.7%   | Pass   |
| Loss on drying             | Gravimetric        | < 0.5%       | 0.21%   | Pass   |
| Residue on ignition        | Gravimetric        | < 0.1%       | 0.03%   | Pass   |
| Heavy metals               | ICP-MS             | < 10 ppm     | < 1     | Pass   |

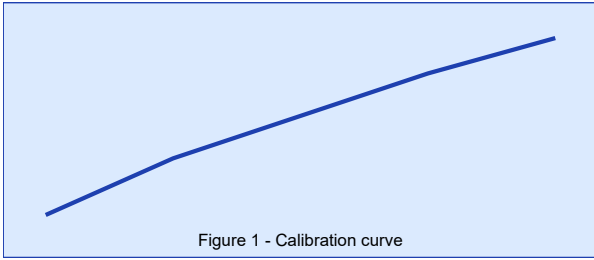


Figure 1 — Instrument calibration curve (data-URI image)

Keep-together paragraph A. These two paragraphs must never be split across a page boundary because the layout engine honours break-inside: avoid on their container.

Keep-together paragraph B. If pagination places paragraph A on one page, paragraph B must appear on the same page or the whole block must move to the next page.

## 1.2 Records

Records paragraph after the keep-together block, used to observe whether the avoid-break block behaves sensibly when followed by more content near the page boundary.

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2. Landscape Section

This section must render as A4 landscape with its own header and footer rules.

| Equipment    | Id    | Range          | Cal due    | Std          | Traceability | Location | Status |
|--------------|-------|----------------|------------|--------------|--------------|----------|--------|
| Balance      | EQ-01 | 0–220 g        | 2026-11-01 | Class E2     | NPL          | Lab 1    | OK     |
| Pipette      | EQ-07 | 1–10 mL        | 2026-12-15 | Gravimetric  | NPL          | Lab 1    | OK     |
| Oven         | EQ-11 | ambient–300 °C | 2027-01-20 | PT-100 probe | NPL          | Lab 2    | OK     |
| pH meter     | EQ-03 | 0–14 pH        | 2026-10-30 | Buffer 4/7/9 | NPL          | Lab 1    | OK     |
| Conductivity | EQ-04 | 0–200 mS       | 2026-11-11 | KCl std      | NPL          | Lab 1    | OK     |
| Incubator    | EQ-09 | 20–60 °C       | 2027-02-01 | PT-100 probe | NPL          | Lab 3    | OK     |

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### 3. Back to Portrait

The final section returns to A4 portrait and reuses the running header pattern with "Page X of Y" in the footer.

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